Student Performance Analytics and AI Score Prediction

Data Analytics with AI — Mini Project Report

1. Abstract

This project demonstrates a simple data analytics workflow using a synthetic dataset of 100 student records. It studies the relationship between weekly study hours, attendance, assignment completion, previous score, and final score. A Linear Regression model is trained to estimate final scores and evaluated on a held-out test set.

2. Problem Statement

Students and educators may want to understand which learning-related indicators are associated with academic outcomes. The project demonstrates how descriptive analytics and a basic machine-learning model can help explore this question.

3. Objectives

• Inspect and summarize the dataset.
• Explore correlations with final score.
• Train a Linear Regression model.
• Evaluate predictions using MAE, RMSE, and R^2.
• Save prediction results and charts.

4. Dataset

The file student_performance.csv contains 100 synthetic records. Features are study_hours_per_week, attendance_percent, assignments_completed_percent, and previous_score. The target variable is final_score (0–100). No real personal or student data is used.

5. Tools and Technologies

Python, pandas, Matplotlib, and scikit-learn. The model uses Linear Regression, an 80/20 train-test split, and a fixed random seed for repeatability.

6. Methodology

The script loads the CSV, prints descriptive statistics, computes correlations, splits features and target into training and test data, fits a Linear Regression model, and reports Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R^2). It also generates plots and a CSV of actual versus predicted test scores.

7. How to Run

Install dependencies with `pip install -r requirements.txt`, then run `python student_performance_analysis.py` from the project folder. Results and charts are saved in the outputs folder.

8. Expected Outputs

Console summaries and model metrics; outputs/prediction_results.csv; outputs/study_hours_vs_score.png; outputs/feature_correlations.png. Exact metric values depend on the installed library versions and dataset.

9. Limitations and Ethical Considerations

The data is artificial and small, so results cannot be generalized to real students. Correlation is not causation. Predictions are estimates and must not be used to label, rank, or penalize students. Any real-world use would require representative data, consent, privacy protection, and validation.

10. Future Scope

Future work could use a larger consented dataset, compare multiple models, add cross-validation, improve visualizations, and create a simple dashboard while documenting fairness and privacy risks.

Prepared as an educational demonstration. Dataset is synthetic.